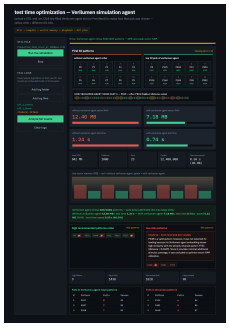
Test Time Optimization

Verilumen Simulation Agent — Dashboard User Guide (2 pages)



This guide explains the dashboard in simple language. The tool compares a **without verilumen agent** (more memory and test time) with **with verilumen agent** (fewer patterns, lower memory and faster test).

How to use the dashboard

Step 1. Upload a STIL file from the left panel.

Step 2. Click **Run live simulation**. **This will take some time to load** — please wait while processing runs. The status line and progress bar show embedding and comparison as results load. Do not move on until the run finishes (status shows **Done**).

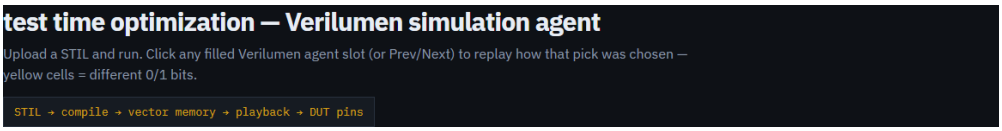
Step 3. Review the first ten picks, memory gauges, and test-time gauges (without vs top 10 pick of verilumen agent).

Step 4. Add fail-log folders from your own computer.

Step 5. Click **Analyze fail counts** for **high recommended patterns order** and **low risk patterns**.

1. Title area

The header shows the product purpose: **test time optimization** with the Verilumen simulation agent. The yellow flow line summarizes the ATE path: STIL → compile → vector memory → playback → DUT pins.



2. Left panel — STIL and fail logs

Use this panel to start work. **Step 1:** upload a **.stil** file. **Step 2:** click **Run live simulation**. **This will take some time to load** — wait for processing while the Verilumen agent embeds patterns and builds the comparison. Watch the status line and progress bar until the run is **Done**. Use **Stop** only if you need to cancel. After the run finishes, add log folders with **Add log folder** (repeat for more folders) and click **Analyze fail counts**. Fail analysis runs locally in the browser — log files stay on your PC.





3. First ten patterns (without vs top 10 pick of verilumen agent)

The left board (**without verilumen agent order**) shows patterns in ordinary order. The right board (**top 10 pick of verilumen agent**) shows diverse patterns the agent chose to **KEEP**. Click slots or use Prev/Next to replay each decision. Yellow-outlined bits are different from the already-kept comparison pattern.

4. Status and progress

The status line explains the current step (embedding, comparing, or done). The yellow progress bar fills as the simulation advances.

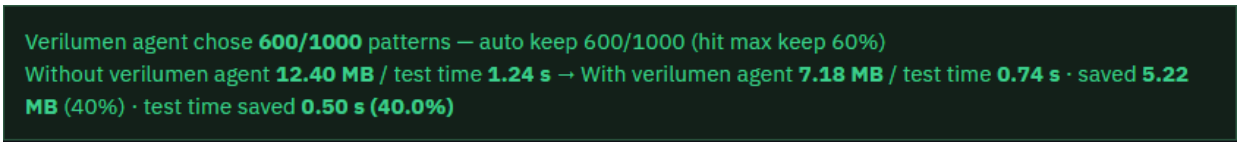
5. Vector memory and test time results

Two red/green gauge pairs show the main optimization results. **without verilumen agent** (red) is the cost of loading/running all patterns. **with verilumen agent** (green) is the cost of the kept subset. Lower green values mean memory and time were saved.



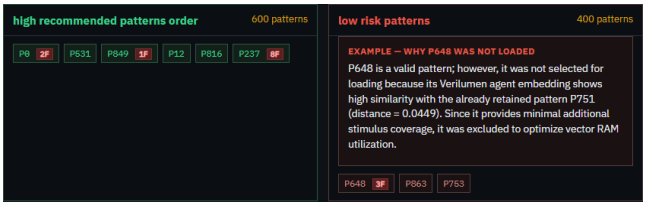
6. Metrics, chart, and final summary

The metrics row shows Host RSS, pattern count, pins, cycles, and test time saved. The live chart plots without-verilumen-agent memory (red) against with-verilumen-agent (green). When the run completes, the green banner reports kept/total patterns, memory, test time, and percentage savings — the main scoreboard for sharing results.

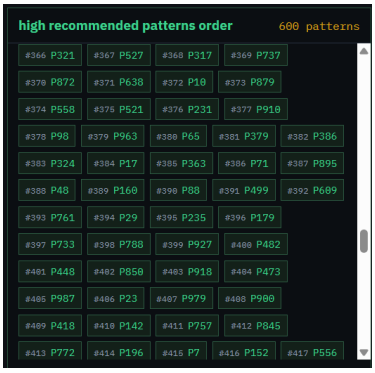


7. High recommended vs low risk pattern lists

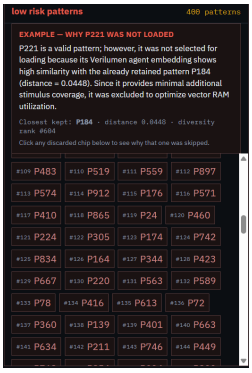
The green list (**high recommended patterns order**) shows patterns loaded into vector memory. The red list (**low risk patterns**) shows patterns skipped to save RAM/time (usually too similar to an already-kept pattern). Click a low-risk chip to read why it was not loaded. After fail analysis, badges like **8F** mean that pattern had 8 fails in your logs.



High recommended patterns order — full pick-order list of kept patterns (scroll to review every loaded id).

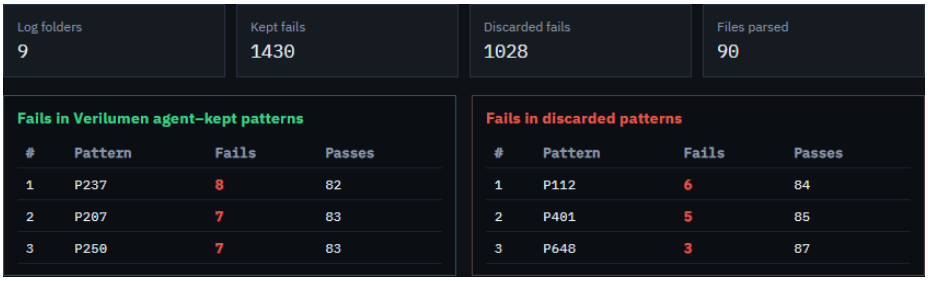


Low risk patterns — skipped patterns; click a chip to see why it was not loaded (closest kept pattern and distance).



8. Fail count analysis

After you add logs and click **Analyze fail counts**, summary cards show folder count, kept fails, discarded fails, and files parsed. Two tables list patterns with the most fails — left **high recommended patterns fails**, right **low risk patterns fail** — sorted highest-first so you can judge coverage risk quickly.



Reading the colors & practical notes

Green = Verilumen / kept / savings. **Red** = full suite / discarded / fails. **Yellow** = progress and bit differences. Always run the simulation before analyzing logs. Expected log style uses pattern headers such as P12 and STATUS:F / STATUS:P. Log labels P1…P1000 map to dashboard ids 0…999 automatically. Live URL: https://vector-memory-optimization.onrender.com